EXERCISE 2

Abel's Formula • Wronskian Dichotomy

Abel's Identity and the Wronskian Dichotomy

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PHASE 1: STATEMENT & SETUP

> Show that the Wronskian for two solutions $x_1(t)$ and $x_2(t)$ of the second order differential equation
>
> $$\frac{d^2x}{dt^2} + p_1(t)\frac{dx}{dt} + p_2(t)x = 0$$
>
> satisfies $\frac{dW(t)}{dt} = -p_1(t)W(t)$. Deduce either that $W(t) = 0$ for all t, or that $W(t) \neq 0$ for all t.

PHASE 2: FUNDAMENTAL FORMULATION

Assume $p_1(t), p_2(t)$ are continuous functions on an interval $I \subseteq \mathbb{R}$. Let $x_1(t), x_2(t) \in C^2(I)$ be solutions of $L[x] = \ddot{x} + p_1\dot{x} + p_2x = 0$. We compute the time derivative of $W(t) = x_1\dot{x}_2 - x_2\dot{x}_1$ and replace the second derivatives using the governing ODE.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

Step 1: Differentiating $W(t)$

$$\frac{dW}{dt} = \frac{d}{dt}\left(x_1\dot{x}_2 - x_2\dot{x}_1\right) = \left(\dot{x}_1\dot{x}_2 + x_1\ddot{x}_2\right) - \left(\dot{x}_2\dot{x}_1 + x_2\ddot{x}_1\right)$$

The product terms $\dot{x}_1\dot{x}_2$ and $\dot{x}_2\dot{x}_1$ cancel:

$$\frac{dW}{dt} = x_1\ddot{x}_2 - x_2\ddot{x}_1 \tag{1}$$

Step 2: Substituting Second Derivatives from the ODE

$$\ddot{x}_1 = -p_1(t)\dot{x}_1 - p_2(t)x_1, \qquad \ddot{x}_2 = -p_1(t)\dot{x}_2 - p_2(t)x_2$$

Substituting into (1):

$$\frac{dW}{dt} = x_1\left(-p_1\dot{x}_2 - p_2x_2\right) - x_2\left(-p_1\dot{x}_1 - p_2x_1\right)$$

$$\frac{dW}{dt} = -p_1(t)\left[x_1\dot{x}_2 - x_2\dot{x}_1\right] - p_2(t)\left[x_1x_2 - x_2x_1\right]$$

Since $x_1x_2 - x_2x_1 \equiv 0$ and $x_1\dot{x}_2 - x_2\dot{x}_1 = W(t)$:

$$\mathbf{\frac{dW(t)}{dt} = -p_1(t)W(t)} \tag{2}$$

Step 3: Integrating to Abel's Formula

Equation (2) is a first-order linear ODE. We do not divide by W (it may vanish, and the dichotomy is deduced afterwards), so we use the integrating factor $\mu(t) = \exp\left(\int_{t_0}^{t} p_1(s)\,ds\right)$. By the Fundamental Theorem of Calculus and the chain rule $\frac{d}{dt}e^u = e^u u'$ with $u(t) = \int_{t_0}^{t} p_1(s)\,ds$:

$$\mu'(t) = p_1(t)\mu(t), \qquad \mu(t_0) = 1, \qquad \mu(t) > 0$$

Using (2) in the product rule:

$$\frac{d}{dt}\left[\mu W\right] = \mu'W + \mu W' = p_1\mu W - p_1\mu W = 0$$

Integrating from t_0 to t (Barrow's rule): $\mu(t)W(t) - \mu(t_0)W(t_0) = \int_{t_0}^{t} 0\,ds = 0$, hence $\mu(t)W(t) = W(t_0)$ and, dividing by $\mu(t) > 0$, for every value of $W(t_0)$ (including 0):

$$\mathbf{W(t) = \frac{W(t_0)}{\mu(t)} = W(t_0)\exp\left(-\int_{t_0}^{t} p_1(s)\,ds\right)} \tag{3}$$

Step 4: Deduction of the Wronskian Dichotomy

Because $p_1(s)$ is continuous on I, the integral $\int_{t_0}^{t} p_1(s)\,ds$ is finite for all $t \in I$. Consequently, the exponential term is strictly positive:

$$\exp\left(-\int_{t_0}^{t} p_1(s)\,ds\right) > 0 \quad \forall t \in I$$

- **If $W(t_0) = 0$:** $W(t) = 0 \times \exp(\dots) = 0$ for all $t \in I$.
- **If $W(t_0) \neq 0$:** Since neither factor is zero, $W(t) \neq 0$ for all $t \in I$.

PHASE 4: INTERPRETATION & MASTER RESULTS

ABEL'S IDENTITY & WRONSKIAN DICHOTOMY

$$\boxed{\frac{dW}{dt} = -p_1(t)W(t) \implies W(t) = W(t_0)\exp\left(-\int_{t_0}^{t} p_1(s)\,ds\right)}$$

Conclusion: For any two solutions of a linear homogeneous second-order ODE with continuous coefficients, the Wronskian is *either identically zero everywhere on I, or non-zero at every point of I*. It can never cross zero isolatedly.